Donghwi Jung

Samsung Electronics
Future Robotics Office - Humanoid Team
Ph.D., Seoul National University, 2026
donghwijung@snu.ac.kr

Research Interests

Robotics, SLAM, Mapping, Localization, Vision-Language Navigation, Human-Robot Collaboration, Bio-inspired Grounding for Embodied Intelligence

Education

Seoul National University Sep. 2020 - Feb. 2026
Ph.D. in Civil & Environmental Engineering
Advisor: Prof. Seong-Woo Kim
Dissertation: "Spatial and Linguistic Understanding for Autonomous Navigation in Smart Cities"

Seoul National University Mar. 2008 - Feb. 2017
B.S. in Food Science & Biotechnology / B.A. in Business Administration

Publications

Donghwi Jung, Jihun Moon, Jiyang Lee, DongYeop Shin, Jinhee Kim, Mingyu Kim, Jungjae Lee, Seong-Woo Kim, *"Beyond Destinations: Instruction-Aware Graph Path Planning and Navigation with OpenStreetMap,"* IEEE Robotics and Automation Letters (RA-L), 2026. (under review)

Donghwi Jung, Keonwoo Kim, Seong-Woo Kim, *"GOTPR: General Outdoor Text-based Place Recognition Using Scene Graph Retrieval with OpenStreetMap,"* IEEE Robotics and Automation Letters (RA-L), May 2025.

Chan Kim*, Keonwoo Kim*, Mintaek Oh, Hanbi Baek, Jiyang Lee, **Donghwi Jung**, Soojin Woo, Younkyung Woo, John Tucker, Roya Firoozi, Seung-Woo Seo, Mac Schwager, and Seong-Woo Kim, *"E2Map: Experience-and-Emotion Map for Self-Reflective Robot Navigation with Language Models,"* 2025 IEEE International Conference on Robotics and Automation (ICRA), Atlanta, USA, May 2025.

Donghwi Jung, Andres Pulido, Jane Shin, Seong-Woo Kim, *"Point Cloud Structural Similarity-Based Underwater Sonar Loop Detection,"* IEEE Robotics and Automation Letters (RA-L), March 2025.

Donghwi Jung, Jae-Kyung Cho, Younghwa Jung, Soohyun Shin, Seong-Woo Kim, *"LoRCoN-LO: Long-term Recurrent Convolutional Network-based LiDAR Odometry,"* International Conference on Electronics, Information, and Communication (ICEIC), Singapore, February 2023.

Donghwi Jung, Younghwa Jung, Woo-Sol Hur, Jae-Kyung Cho, and Seong-Woo Kim, *"Multi-Floor Mapping including the Elevator,"* Summer Annual Conference of the Institute of Electronics and Information Engineers, Jeju, Republic of Korea, June 2022.

Byeong Yeon Ryu, Won Nyoung Park, **Donghwi Jung** and Seong-Woo Kim, *"Landmark Localization for Drone Aerial Mapping Using GPS and Sparse Point Cloud for Photogrammetry Pipeline Automation,"* International Conference on Electronics, Information, and Communication (ICEIC), Jeju, Republic of Korea, February. 2022.

Projects

Perceptive Humanoid Locomotion Control Mar. 2026 – Present
Internal Project
Samsung Electronics Future Robotics Office
Role – Methodology Development, Environment Setup, Model Training, and Experimental Validation

Visual-Language Outdoor Navigation Using OpenStreetMap Apr. 2025 - Sep. 2025
Internal Laboratory Project
Role - Navigation Algorithm Implementation, Data Acquisition and Experimentation, Team Leadership

Visual-Language Indoor Navigation Using Foundation Models Apr. 2024 - Sep. 2024
Internal Laboratory Project
Role - 2D/3D Mapping, Virtual Environment Implementation, Visual-Language Map Generation, and Localization

Development of strong autonomous intelligence for unmanned vehicles Jan. 2022 - July 2023
R&D Project
funds from the Agency for Defense Development, Republic of Korea
Role - LiDAR Odometry Estimation

Location estimation method in pretreatment and painting factory using inertial sensors Sep. 2022 - Feb. 2023
Industrial Cooperative Project
funds from Daewoo Shipbuilding & Marine Engineering Co., Ltd
Role - Inertial Odometry Estimation

Yard 3D map creation and virtual environment establishment Apr. 2022 - Oct. 2022
Industrial Cooperative Project
funds from Daewoo Shipbuilding & Marine Engineering Co., Ltd
Role - Data Acquisition, Virtual Environment Implementation, 3D Mapping

Development of AI-based transporter safety driving technology Sep. 2021 - Sep. 2022
Industrial Cooperative Project
funds from Daewoo Shipbuilding & Marine Engineering Co., Ltd
Role - Data Acquisition, Mapping, and Object Detection

Teaching Experience

Teaching Assistant Fall. 2021 - Spring. 2022
How to Make a Robot with Artificial Intelligence
College of Engineering, Seoul National University

Teaching Assistant Summer. 2021 - Spring. 2022
Basic Computing: First Adventures in Computing
Faculty of Liberal Education, Seoul National University

Teaching Assistant Spring. 2021
Basic Mathematics and Programming Practice for Machine Learning
College of Engineering, Seoul National University

Global Experience

Visiting Student Jul. 2023 – Feb. 2024

Active Perception and Robot Intelligence Lab, University of Florida
Advisor: Prof. Jane Shin
Research on sonar-based underwater image super-resolution and point cloud loop detection

Exchange Student Jul. 2013 – May. 2014
University of Singapore

Presentation

IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS) Oct. 22, 2025
Hangzhou, China
Title – GOTPR: General Outdoor Text-based Place Recognition Using Scene Graph Retrieval with OpenStreetMap

IEEE International Conference on Robotics and Automation (ICRA) May 22, 2025
Atlanta, USA
Title – E2Map: Experience and Emotion Map for Self-Reflective Robot Navigation with Language Models

The 18th KICS Future Communication Technology Workshop May 23, 2025
Seoul, Republic of Korea
Title – Language Understanding for Spatial Understanding: Human–Robot Communication

Awards

Best Paper Award (Bronze Prize) Feb. 2023
International Conference on Electronics, Information, and Communication (ICEIC)
Title – LoRCoN-LO: Long-Term Recurrent Convolutional Network-Based LiDAR Odometry

Outstanding Student Paper Award Jun. 2022
Summer Annual Conference of the Institute of Electronics and Information Engineers (IEIE)
Title – Multi-Floor Mapping including the Elevator

Work Experience

Samsung Electronics Seoul, Republic of Korea
Future Robotics Office Mar. 2026 - Present
Staff Engineer
Developing perceptive locomotion control algorithms for humanoid robots.

Skills

Programming Skills
ROS (Robot Operating System) 1 and 2, C++, Python, PyTorch, OpenCV, g2o, Unity

Sensors
LiDAR, Camera, IMU, GPS, Wheel Encoder

Platforms
Quadruped robot, Mobile robot, Autonomous vehicle

URLS

- Website (https://donghwijung.github.io)
- Github (https://github.com/donghwijung)
- Youtube (https://www.youtube.com/channel/UCqU4Gn7X3WlivtAI_yRaXOw)
- LinkedIn (https://www.linkedin.com/in/donghwi-jung-6b9057a9/)